

RADHEY KALRA

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Open source contributor and full-stack developer working across ML/audio systems, Linux tooling, and self-hosted infrastructure.

EDUCATION

Bachelor of Technology in CSE

Model Institute of Engineering and Technology

Aug. 2023 – May 2027

Jammu and Kashmir, India

Schooling (Class I – XII)

Air Force School, Jammu

2011 – 2023

Jammu and Kashmir, India

PROJECTS

BoxBox

Nov. 2025 – Present

Self-hosted file management system for Linux servers

- Built a **lightweight** self-hosted file manager in **Svelte + Go** with multi-mount browsing, chunked transfers, real-time updates, media preview/stream/download support, and Monaco Editor integration.
- Shipped production deployments as a **31MB single Go binary** with an embedded Svelte frontend and Docker-based setup.
- Grew the project to **200+ GitHub stars** and **4K+ downloads**.

U-Net Audio Filter

Jul. 2025 – Present

Audio enhancement by removing background noise

- Building an audio enhancement system with a **custom U-Net-inspired architecture** and end-to-end audio denoising pipeline.
- Implemented **Python/PyTorch + audio DSP** preprocessing, spectrogram modeling, and evaluation tooling, with early results showing minor gains over open-source baselines.

HPomen Linux (Victus Control)

Mar. 2026 – Jul. 2026

Reverse-engineering HP OMEN control utility behavior on Linux

- Built a Linux-first OMEN/Victus control suite in **Vala** with a GTK4 monitor app, tray companion, D-Bus helper, and probe CLI.
- Reverse-engineered **HP WMI/sysfs** behavior to validate profile/fan controls (**auto/max**) and extend **hp-wmi** with capability-gated safety fallbacks.
- Submitted a Linux kernel **hp-wmi** patch accepted into **platform-driver-x86**, adding Victus 15-fb0xxx board support for fan-control feature data; also provided **Tested-by** validation for another **hp-wmi** fan-control patch on real hardware.

Fast Blob Storage (FBS)

Apr. 2026 – May 2026

S3-compatible blob storage and CDN solution primarily for self-hosting

- Built a **Go** storage backend with **SQLite** metadata, **SvelteKit** dashboard, and a published Docker image.
- Reached **50+ GitHub stars** while supporting self-hosted S3-compatible object storage and CDN workflows.
- Supported Both SigV4 and bearer token authentication, using linux **sendfile()** for efficient zero-copy file transfers.
- Achieved binary sizes of **15MB** and idel RSS of **16MB** and under load RSS of **21MB** in controled benchmarks, making it suitable for self-hosting on resource-constrained environments.

Additional Projects

- **QckPages**: SvelteKit/Convex app for publishing HTML pages, FBS-backed storage.
- **rsclip**: Rust Wayland clipboard manager with a low-memory daemon, GTK4 UI, SQLite history, search, filters, OCR Support and Secrets management.
- **Dots-Niri**: Replicates my NiriWM rice on Arch-based install with a single **curl** command.
- **HowTo**: Astro tutorials site for practical hardware and software walkthroughs.
- **Pico32**: C++/PlatformIO serial TUI built for ESP32 Wi-Fi and Bluetooth pentesting workflows.
- **TL;DR**: Electron app that summarizes WhatsApp chats with OpenRouter compatible AI models.
- **Hotel Booking**: Svelte/Express/MySQL booking platform with authentication and admin workflows.
- **Tenant Manager**: React Native app for managing tenants and tracking electricity usage.

CO-CURRICULAR ACTIVITIES

Cloud Lead

Aug. 2025 – Present

GDGoC (Google Developer Groups on Campus), MIET Jammu

Jammu and Kashmir, India

- Led cloud learning initiatives through hands-on **Google Cloud** workshops and guided sessions for **100+ students**, and built a tutorial platform with **20+** lab walkthroughs, notes, and video resources to support **Google Cloud Study Jam** completion.

Cloud and DevOps Lead

Jan. 2026 – Present

AWS Cloud Club MIET Jammu

Jammu and Kashmir, India

- Lead cloud and DevOps initiatives through hands-on workshops, technical sessions, and collaborative projects, driving adoption of **AWS**, containerization, and infrastructure best practices focused on deployment automation and scalability.
- IDE Bootcamp Edition 3 Team Leader – 2nd Place**

Apr. 2026

Innovation, Design, and Entrepreneurship (IDE) Bootcamp

GNDEC, Ludhiana, Punjab
- Led a team to secure **2nd Place** at the AICTE/MoE Innovation Cell bootcamp by applying rapid prototyping and business modeling to develop an AI-powered B2B SaaS product for automated presentation generation from user-uploaded templates and documents.
- IDE Bootcamp Edition 2 Team Leader**

Feb. 2025

Innovation, Design, and Entrepreneurship (IDE) Bootcamp

IIT Jammu
- Participated in a 5-day immersive bootcamp at IIT Jammu with Wadhvani Foundation, AICTE, and MoE’s Innovation Cell, gaining business modeling, product design, and entrepreneurship experience while leading a team to develop a “Sports Match Finder” app using design thinking and the Lean Canvas model.
- Hackathon Team Leader**

Internal Hackathon & Smart India Hackathon (SIH 2024 & 2025)

MIET, Jammu
- Led multidisciplinary teams in MIET’s internal hackathon and Smart India Hackathon selection rounds in **2024** and **2025**, reaching the **top 9 teams in the college** both years through collaborative problem-solving, rapid prototyping, and real-world solution design.

TECHNICAL SKILLS

Languages: Python, C/C++, TypeScript, Go, Rust, SQL, Bash, PowerShell, QML, Vala
Frameworks & Platforms: SvelteKit, React, Astro, Tailwind CSS, React Native, Bun
Developer Tools: Git, Docker, CI/CD, Cloudflare, Vercel, Kubernetes, Terraform
Databases & Cloud: Convex, PostgreSQL, MySQL, AWS, GCP
Domains: Audio Processing, Linux Servers, Full-Stack Development

CERTIFICATES

Cybersecurity Coursera: Google Cybersecurity; Cisco Cybersecurity Operations; EC-Council Cybersecurity Attack and Defense Fundamentals; Palo Alto Networks Cybersecurity. NPTEL: Ethical Hacking (IIT Kharagpur)	AI NPTEL: Deep Learning (IIT Ropar); Artificial Intelligence for Management. IBM SkillsBuild: AI Agent Architect. Coursera: ServiceNow AI-Powered Workflows
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